

Text Mining

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Introduction

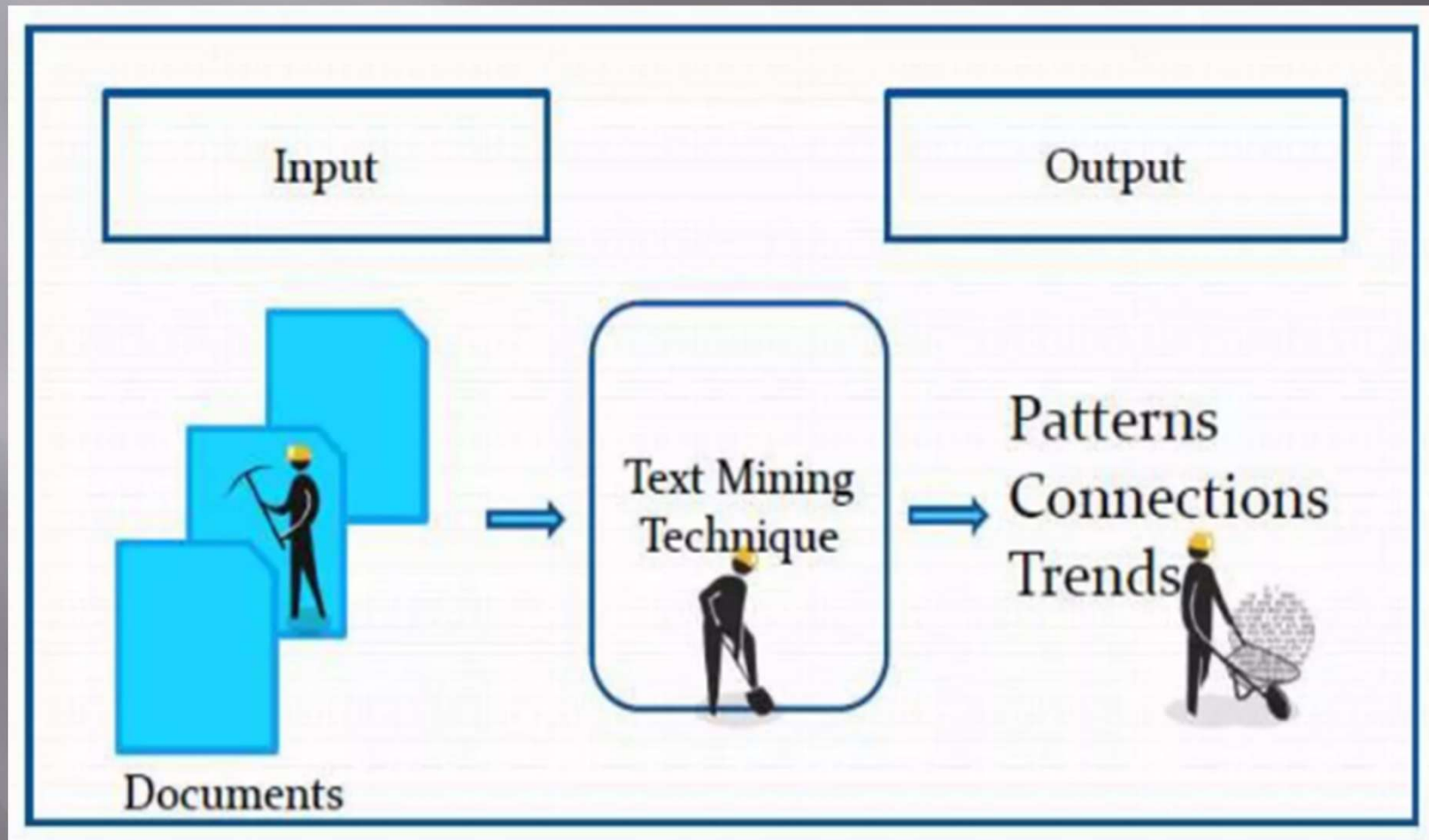
- ❖ It is the process of **extracting required data** and consists of large collections of documents from various sources.
- ❖ Ex: News articles, research papers, digital libraries, etc.
- ❖ The main goal is this concept is to **find patterns** and trends across multiple documents.



Data Mining Vs Text Mining

Data Mining	Text Mining
Process directly	Linguistic processing or natural language processing (NLP)
Identify causal relationship	Discover heretofore unknown information
Structured Data	Semi-structured & Unstructured Data (Text)
Structured numeric transaction data residing in rational data warehouse	Applications deal with much more diverse and eclectic collections of systems and formats

Input-Output model for Text Mining



Text Mining: Process

Text Pre-processing:

- ❖ It is the process of extracting useful information and knowledge from the unstructured text data.
- ❖ It includes methods like cleaning and tokenization.

Feature Selection:

- ❖ The process of reducing the input of processing or finding the essential information sources.

Evaluate:

- ❖ Computational linguistic principles are used to evaluate text.
- ❖ The principles like Part-of-Speech Tagging and Syntax and Parsing .

It consists of

- a. Text Data Analysis and Information Retrieval
 - b. Dimensionality Reduction for text
 - c. Text Mining Approaches
-
- a) Text Data Analysis and IR: Organization and retrieval of information from large number of text documents DBMS (concurrency control, Recovery management, transaction management) vs. IR (Unstructured, Approximation, relevance)

b) Dimensionality Reduction for text

- ▣ Dimensionality reduction technique can be defined as, *"It is a way of converting the higher dimensions dataset into lesser dimensions dataset ensuring that it provides similar information."* These techniques are widely used in [machine learning](#) for obtaining a better fit predictive model while solving the classification and regression problems.
- ▣ It is commonly used in the fields that deal with high-dimensional data, such as **speech recognition, signal processing, bioinformatics, etc.** It can also be used for data visualization, noise reduction, cluster analysis, etc.

c) Text mining approaches

- ▣ **Keyword based**
- ▣ **Tagging based** : Automatic tagging
- ▣ **Information extraction based**: Sematic information, NPL(Natural Processing Language, Machine Learning methods...)
- i) **Keyword-Based Association Analysis**: Motivation is the collect sets of keywords or terms that occur frequently together and then find the association or correlation relationships among them.
 - The number of meaningless results and the execution time is greatly reduced.

ii) Tagging based: Automatic tagging:

- ▣ Document Classification Analysis: The Motivation is the automatic classification for the large number of online text documents(web pages, e-mails, etc).
- ▣ Classification Process: Data preprocessing
 - Definition of training set and test sets
 - Creation of the classification model using the selected classification algorithm
 - Classification model validation
 - Classification of new/unknown text data.

Text Retrieval Methods:

- ▣ Text queries -> Vectors -> search for nearer neighbours in “document collection”.
- ▣ Problems: Synonymy (Term referring same meaning(EX: near, close), Polysemy(same word different meanings(EX: mining may be data mining, land or coal mining))....
- ▣ Solutions:
 - a) Latent semantic indexing
 - b) Reduce Dimensionality of documents, uses SVD
 - c) Locality Preserving indexing
 - d) Discriminative features, Mapping semantics, similarity matrix
 - e) Probability Latent Semantic analysis
 - f) Mixture model with k latent common themes as components

Text Retrieval Methods

- ✓ Term Frequency Matrix: $TF(d, t)$ measures association of term 't' with respect to given document 'd'.

$$TF(d, t) = \begin{cases} 0 & \text{if } freq(d, t) = 0 \\ 1 + \log(1 + \log(freq(d, t))) & \text{otherwise.} \end{cases}$$

$$IDF(t) = \log \frac{1 + |d|}{|d_t|}$$

$$TF-IDF(d, t) = TF(d, t) \times IDF(t)$$

Example:

A term frequency matrix showing the frequency of terms per document.

document/term	t_1	t_2	t_3	t_4	t_5	t_6	t_7
d_1	0	4	10	8	0	5	0
d_2	5	19	7	16	0	0	32
d_3	15	0	0	4	9	0	17
d_4	22	3	12	0	5	15	0
d_5	0	7	0	9	2	4	12

✓ Calculate the TF-IDF value of a term in a document ?

$$TF(d_4, t_6) = 1 + \log(1 + \log(15)) = 1.3377$$

$$IDF(t_6) = \log \frac{1+5}{3} = 0.301.$$

$$TF-IDF(d_4, t_6) = 1.3377 \times 0.301 = 0.403$$

Applications

1. News Article Classification
2. Automatic Email Filtering
3. Webpage Classification
4. Word Sense Disambiguation..

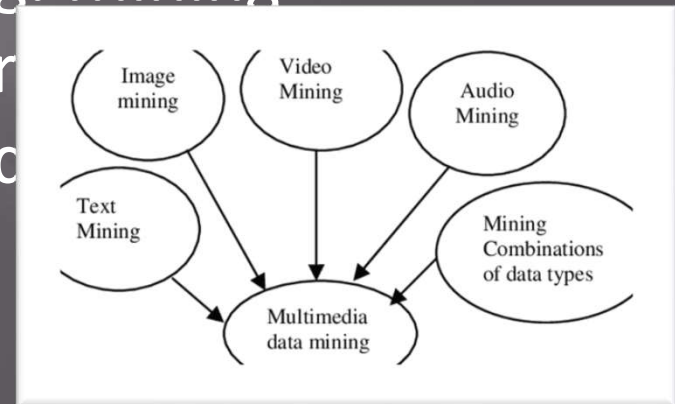
MULTIMEDIA DATA MINING

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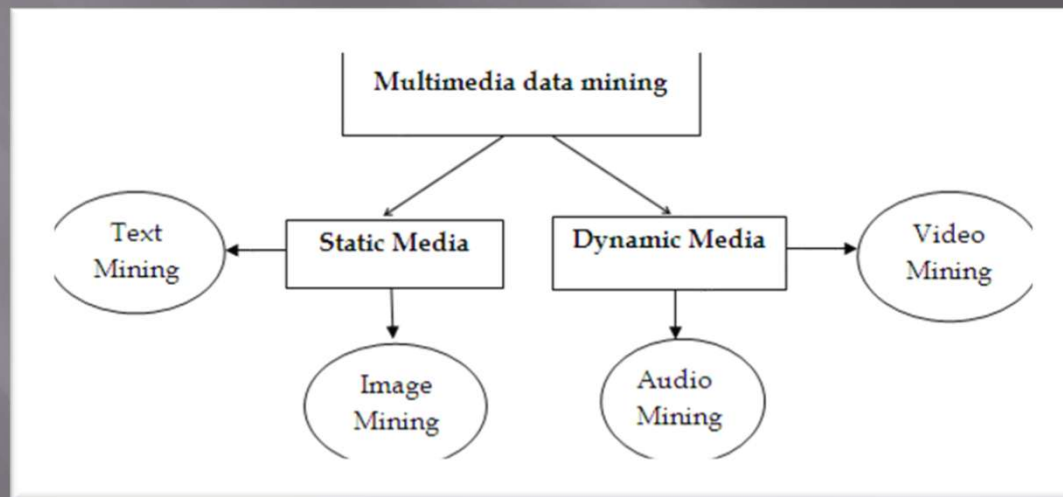
What is Multimedia Data Mining?

- ▣ Multimedia mining is a subfield of data mining that is used to find interesting information of implicit knowledge from multimedia databases. Mining in multimedia is referred to as automatic annotation or annotation mining. Mining multimedia data requires two or more types of data, such as text and video or text and video.



Types of Multimedia Applications

1. Repository applications
2. Presentation applications
3. Collaborative work using multimedia information



1.Repository applications: A Large amount of multimedia data and meta-data (Media format data, Media keyword data, Media feature data) that is stored for retrieval purposes, e.g., Repository of satellite images, engineering drawings, radiology scanned pictures.

2.Presentation applications: They involve delivering multimedia data subject to temporal constraints. Optimal viewing or listening requires DBMS to deliver data at a certain rate, offering the quality of service above a certain threshold. Here data is processed as it is delivered. Example: Annotating of video and audio data, real-time editing analysis.

3.Collaborative work using multimedia information involves executing a complex task by merging drawings and changing notifications. Example: Intelligent healthcare network.

Challenges with Multimedia Database



1.Modelling – Working in this area can improve database versus information retrieval techniques thus, documents constitute a specialized area and deserve special consideration.

2.Design – The conceptual, logical and physical design of multimedia databases has not yet been addressed fully as performance and tuning issues at each level are far more complex as they consist of a variety of formats like JPEG, GIF, PNG, MPEG which is not easy to convert from one form to another.

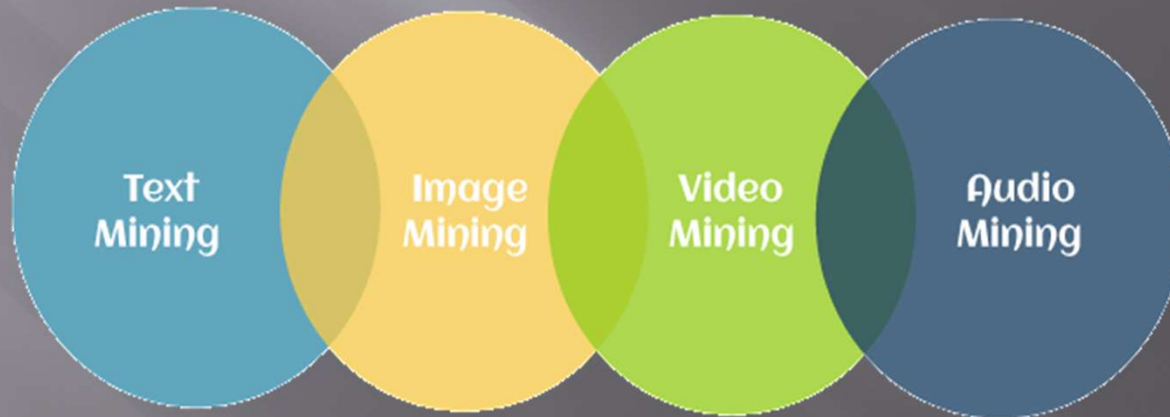
3.Storage – Storage of multimedia database on any standard disk presents the problem of representation, compression, mapping to device hierarchies, archiving and buffering during input-output operation. In DBMS, a "BLOB"(Binary Large Object) facility allows untyped bitmaps to be stored and retrieved.

4. Performance - For an application involving video playback or audio-video synchronization, physical limitations dominate. The use of parallel processing may alleviate some problems but such techniques are not yet fully developed. Apart from this multimedia database consume a lot of processing time as well as bandwidth.

5. Queries and retrieval -For multimedia data like images, video, audio accessing data through query opens up many issues like efficient query formulation, query execution and optimization which need to be worked upon.

Categories of Multimedia Data Mining

Categories of Multimedia Data Mining



1. Text Mining

Text is the foremost general medium for the proper exchange of information. Text Mining evaluates a huge amount of usual language text and detects exact patterns to find useful information.

2. Image Mining

Image mining systems can discover meaningful information or image patterns from a huge collection of images. It includes digital image processing, image understanding, database, AI, etc.

3. Video Mining

Video mining is unsubstantiated to find interesting patterns from many video data; multimedia data is video data such as text, image, metadata, visuals and audio. It is commonly used in security and surveillance, entertainment, medicine, sports and education programs.

4. Audio Mining

Audio mining plays an important role in multimedia applications, is a technique by which the content of an audio signal can be automatically searched, analyzed and rotten with wavelet transformation . Band energy, frequency centroid, zero-crossing rate, pitch period and bandwidth are often used for audio processing.

Application of Multimedia Mining



•**Digital Library:** The collection of digital data is stored and maintained in a digital library, which is essential to convert different digital data formats into text, images, video, audio, etc.

•**Traffic Video Sequences:** To determine important but previously unidentified knowledge from the traffic video sequences, detailed analysis and mining are to be performed based on vehicle identification, traffic flow, and queue temporal relations of the vehicle at an intersection. This provides an economic approach for regular traffic monitoring processes.

•**Medical Analysis:** Multimedia mining is primarily used in the medical field, particularly for analyzing medical images. Various data mining techniques are used for image classification. Examples, Automatic 3D delineation of highly aggressive brain tumours, Automatic localization and identification of vertebrae in 3D CT scans, MRI Scans, ECG and X-Ray.

Customer Perception: It contains details about customers' opinions, products or services, customers complaints, customers preferences, and the level of customer satisfaction with products or services, which are collected together. The audio data serve as topic detection, resource assignment and evaluation of the quality of services. Many companies have call centres that receive telephone calls from customers.

•**Media Making and Broadcasting:** Radio stations and TV channels create broadcasting companies, and multimedia mining can be applied to monitor their content to search for more efficient approaches and improve their quality.

Thank you

SEQUENTIAL PATTERN MINNING ALGORITHMS

CONTENTS

- ▣ SPADE Algorithm implementation
- ▣ PREFIX SPAN Algorithm implementation
- ▣ Mining multi dimensional,multi level sequential patterns
- ▣ Applications sequence pattern mining algorithms

Sequential database

- ▣ A sequential database consists of ordered elements or events.
- ▣ TRANSACTIONAL DATABASE
SEQUENCE DATABASE

TID	ITEM SETS
1	a,b,d
2	a,,c,d
3	a,d,e
4	b,e,f

SID	SEQUENCE
1	<a(abc)(ac)d(cf)>
2	<(ad)c(bc)(ac)>
3	<(ef)(ab)(df)cb>
4	<(ef)(ab)(df)cb>

SPADE ALGORITHM

- ▣ SPADE(Sequential Pattern Discovery using Equivalent class)developed by Zaki 2001
- ▣ A vertical format sequential pattern mining method
- ▣ A sequence database is mapped to a large set of items $\langle \text{SID}, \text{EID} \rangle$
- ▣ Sequential pattern mining is performed by growing the sub sequences(patterns)one item at a time by Apriori candidate generation

- ▣ In spade algorithm we will generate vertical data format from the horizontal data format

A sequence database

Sequence_ID	Sequence
1	$\langle a(abc)(ac)d(cf) \rangle$
2	$\langle (ad)c(bc)(ae) \rangle$
3	$\langle (ef)(ab)(df)cb \rangle$
4	$\langle eg(af)cbc \rangle$

Finding the patterns starting with $\langle\{a,b\}\rangle$

PrefixSpan does a database projection with $\langle\{a,b\}\rangle$:

Projected database of $\langle\{a\}\rangle$

$S_1 =$	$\langle\{_b\}, \{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{_b\}, \{b\}, \{c\}\rangle$
$S_4 =$	$\langle\{_b\}, \{c\}\rangle$



Projected database of $\langle\{a,b\}\rangle$

$S_1 =$	$\langle\{_b\}, \{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{_b\}, \{b\}, \{c\}\rangle$
$S_4 =$	$\langle\{_b\}, \{c\}\rangle$

Result:

$\langle\{a,b\}, \{c\}\rangle$ support : 3
 $\langle\{a,b\}, \{a\}\rangle$ support : 1
 $\langle\{a,b\}, \{b\}\rangle$ support : 1



Projected database of $\langle\{a,b\}\rangle$

$S_1 =$	$\langle\{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{b\}, \{c\}\rangle$
$S_4 =$	$\langle\{c\}\rangle$

Vertical Database Format

SID	EID	Items
1	1	a
1	2	abc
1	3	ac
1	4	d
1	5	cf
2	1	ad
2	2	c
2	3	bc
2	4	ae
3	1	ef
3	2	ab
3	3	df
3	4	c
3	5	b
4	1	e
4	2	g
4	3	af
4	4	c
4	5	b
4	6	c

ID_lists for some 1-sequences

a		b		...
SID	EID	SID	EID	...
1	1	1	2	
1	2	2	3	
1	3	3	2	
2	1	3	5	
2	4	4	5	
3	2			
4	3			

ID_lists for 2-sequences

ab			ba			...
SID	EID (a)	EID(b)	SID	EID (b)	EID(a)	...
1	1	2	1	2	3	
2	1	3	2	3	4	
3	2	5				
4	3	5				

aba				...
SID	EID (a)	EID(b)	EID(a)	...
1	1	2	3	
2	1	3	4	

ID_lists for 3-Sequences

PREFIX SPAN ALGORITHM

- ▣ Proposed by Jian Pei et al(2001)
- ▣ The algorithm is designed to only consider patterns that exist in the database
- ▣ This algorithm uses a concept of database projection and depth first search.
- ▣ This is the not most efficient algorithm, but it is simple and easy to extend ,so it is popular.
- ▣ Taking a small example

s1	$\langle \{a,b\}, \{c\}, \{a\} \rangle$
s2	$\langle \{a,b\}, \{b\}, \{c\} \rangle$
s3	$\langle \{b\}, \{c\}, \{d\} \rangle$
s4	$\langle \{b\}, \{a,b\}, \{c\} \rangle$

Step 2

PrefixSpan eliminates infrequent items:

Sequence database

$S_1 =$	$\langle \{a, b\}, \{c\}, \{a\} \rangle$
$S_2 =$	$\langle \{a, b\}, \{b\}, \{c\} \rangle$
$S_3 =$	$\langle \{b\}, \{c\}, \{d\} \rangle$
$S_4 =$	$\langle \{b\}, \{a, b\}, \{c\} \rangle$

Result:

$\langle \{a\} \rangle$ support : 3
 $\langle \{b\} \rangle$ support : 4
 $\langle \{c\} \rangle$ support : 4
 ~~$\langle \{d\} \rangle$ support : 1~~

$minsup = 3$

an algorithm

Step 3 – Find patterns starting with $\langle\{a\}\rangle$

PrefixSpan does a database projection with $\langle\{a\}\rangle$

Sequence database

$S_1 =$	$\langle\{a, b\}, \{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{a, b\}, \{b\}, \{c\}\rangle$
$S_3 =$	$\langle\{b\}, \{c\}, \{d\}\rangle$
$S_4 =$	$\langle\{b\}, \{a, b\}, \{c\}\rangle$

Projected database of $\langle\{a\}\rangle$

$S_1 =$	$\langle\{\text{a}, b\}, \{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{\text{a}, b\}, \{b\}, \{c\}\rangle$
$S_3 =$	$\langle\{b\}, \{c\}, \{d\}\rangle$
$S_4 =$	$\langle\{b\}, \{a, b\}, \{c\}\rangle$

$minsup = 3$

What is a database projection?

It means to keep only the sequences containing $\langle\{a\}\rangle$.

Moreover, for these sequences, we delete the first occurrence of $\langle\{a\}\rangle$ and everything that appears before.

Result of step 3

Sequence database

$S_1 =$	$\langle \{a, b\}, \{c\}, \{a\} \rangle$
$S_2 =$	$\langle \{a, b\}, \{b\}, \{c\} \rangle$
$S_3 =$	$\langle \{b\}, \{c\}, \{d\} \rangle$
$S_4 =$	$\langle \{b\}, \{a, b\}, \{c\} \rangle$



Projected database of $\langle \{a\} \rangle$

$S_1 =$	$\langle \{ _b \}, \{c\}, \{a\} \rangle$
$S_2 =$	$\langle \{ _b \}, \{b\}, \{c\} \rangle$
$S_4 =$	$\langle \{ _b \}, \{c\} \rangle$

Then, PrefixSpan counts the support of each sequential pattern starting with $\langle \{a\} \rangle$ that has one more item:

Projected database of $\langle \{a\} \rangle$

$S_1 =$	$\langle \{ _b \}, \{c\}, \{a\} \rangle$
$S_2 =$	$\langle \{ _b \}, \{b\}, \{c\} \rangle$
$S_4 =$	$\langle \{ _b \}, \{c\} \rangle$

Result:

$\langle \{a\}, \{a\} \rangle$ support : 1

$\langle \{a\}, \{b\} \rangle$ support : 1

$\langle \{a\}, \{c\} \rangle$ support: 3

$\langle \{a, b\} \rangle$ support : 3

$minsup = 3$

Prefix span then extends each pattern recursively

- Let us start with $\langle\{a\},\{c\}\rangle$

PrefixSpan does a database projection with $\langle\{a\},\{c\}\rangle$:

Projected database of $\langle\{a\}\rangle$

$S_1 =$	$\langle\{_b\},\{c\},\{a\}\rangle$
$S_2 =$	$\langle\{_b\},\{b\},\{c\}\rangle$
$S_4 =$	$\langle\{_b\},\{c\}\rangle$



Projected database of $\langle\{a\},\{c\}\rangle$

$S_1 =$	$\langle\{_b\},\{c\},\{a\}\rangle$
$S_2 =$	$\langle\{_b\},\{b\},\{c\}\rangle$
$S_4 =$	$\langle\{_b\},\{c\}\rangle$

Result:

~~$\langle\{a\},\{c\},\{a\}\rangle$ support: 1~~

Finding the patterns starting with $\langle\{a,b\}\rangle$

PrefixSpan does a database projection with $\langle\{a,b\}\rangle$:

Projected database of $\langle\{a\}\rangle$

$S_1 =$	$\langle\{_b\}, \{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{_b\}, \{b\}, \{c\}\rangle$
$S_4 =$	$\langle\{_b\}, \{c\}\rangle$



Projected database of $\langle\{a,b\}\rangle$

$S_1 =$	$\langle\{_b\}, \{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{_b\}, \{b\}, \{c\}\rangle$
$S_4 =$	$\langle\{_b\}, \{c\}\rangle$

Result:

$\langle\{a,b\}, \{c\}\rangle$ support : 3
 $\langle\{a,b\}, \{a\}\rangle$ support : 1
 $\langle\{a,b\}, \{b\}\rangle$ support : 1



Projected database of $\langle\{a,b\}\rangle$

$S_1 =$	$\langle\{c\}, \{a\}\rangle$
$S_2 =$	$\langle\{b\}, \{c\}\rangle$
$S_4 =$	$\langle\{c\}\rangle$

Finding patterns $\langle \{a, b\}, \{c\} \rangle$

PrefixSpan does a database projection for $\langle \{a, b\}, \{c\} \rangle$:

Projected database of $\langle \{a, b\} \rangle$

$S_1 =$	$\langle \{c\}, \{a\} \rangle$
$S_2 =$	$\langle \{b\}, \{c\} \rangle$
$S_4 =$	$\langle \{c\} \rangle$



Projected database of $\langle \{a, b\}, \{c\} \rangle$

$S_1 =$	$\langle \{a\} \rangle$
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Result:

$\langle \{a, b\}, \{c\}, \{a\} \rangle$ support : 1

This pattern is infrequent!

Now again finding patterns starting with $\langle\{b\}\rangle$

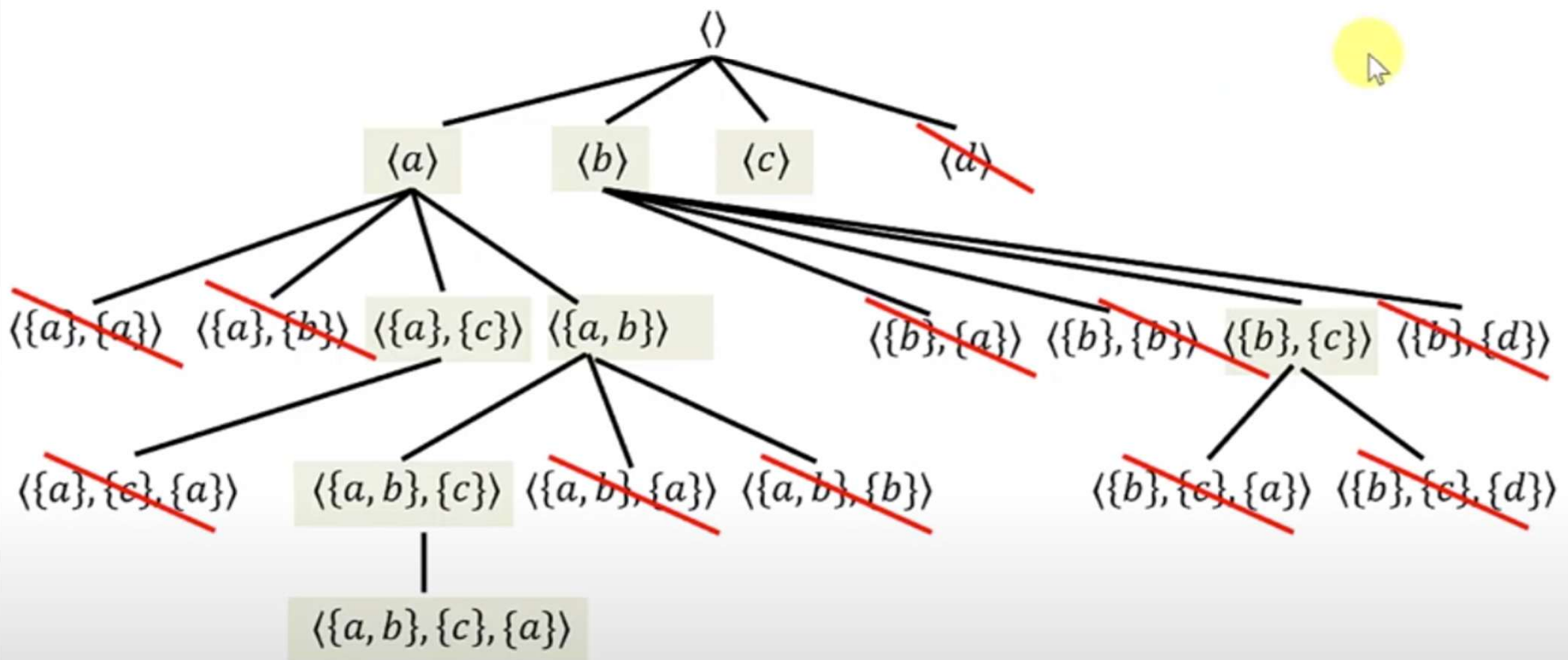
Final result is:

Those are the frequent sequential patterns:

- $\langle\{a\}\rangle$ support : 3
- $\langle\{b\}\rangle$ support : 4
- $\langle\{c\}\rangle$ support : 4
- $\langle\{a\}, \{c\}\rangle$ support: 3
- $\langle\{a, b\}\rangle$ support : 3
- $\langle\{a, b\}, \{c\}\rangle$ support : 3
- $\langle\{b\}, \{c\}\rangle$ support : 3

Final observation

PrefixSpan performs a depth-first search:



APPLICATIONS

- ▣ Customer shopping sequences
- ▣ Purchase a laptop first, then a digital camera, and then a smartphone, with in 6 months.
- ▣ Medical treatments, natural disasters(earth quakes),
- ▣ Science and engineering process and markets.
- ▣ Calling patterns for telephones,
- ▣ Biological sequences : DNA sequence, protein sequences.

SPATIAL DATA MINING

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CONTENTS

- ▣ WHAT IS SPATIAL DATA MINING?
- ▣ TYPES OF SPATIAL DATA
- ▣ STEPS IN SPATIAL MINING ANALYSIS
- ▣ TECHNIQUES USED IN SPATIAL MINING
- ▣ APPLICATIONS
- ▣ CONCLUSION

What is spatial data mining?

- ▣ Spatial data mining is the process of extracting useful information and patterns from spatial data. (or)
- ▣ Spatial data mining is a specialized subfield of data mining that deals with extracting knowledge from spatial data.

SPATIAL DATA

- ▣ Spatial data refers to data that is associated with a particular location or geography.
- ▣ Examples of spatial data include maps, satellite images, GPS data, and other geospatial information

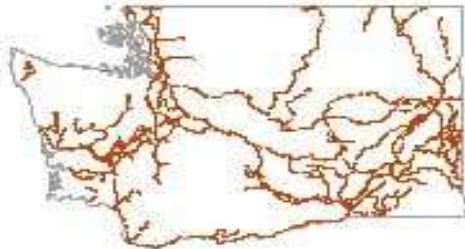
Types of Spatial Data

- Different types of spatial data are used in spatial data mining. These include point data, line data, and polygon data.
- **Point Data**
 - Point data represents a single location or a set of locations on a map. Each point is defined by its x and y coordinates, representing its position in the geographic space. Point data is commonly used to represent geographic features such as cities, landmarks, or specific locations of interest. Examples of point data in transportation include delivery locations, bus stops, or railway stations.
- **Line Data**
 - Line data represents a linear feature, such as a road, a river, or a pipeline, on a map. Each line is defined by a set of vertices, which represent the start and end points of the line. Line data is commonly used to represent transportation networks, such as roads, highways, or railways. Line data is also used in other areas, such as hydrology, geology, or ecology, to represent streams, faults, or animal migration routes.
- **Polygon Data**
 - Polygon data represents a closed shape or an area on a map. Each polygon is defined by a set of vertices that connect to form a closed boundary. Polygon data is commonly used to represent administrative boundaries, land use, or demographic data. In transportation, polygon data can be used to represent areas of interest, such as delivery zones or traffic zones.
- In summary, point data represents a single location, line data represents a linear feature, and polygon data represents an area or a closed shape.

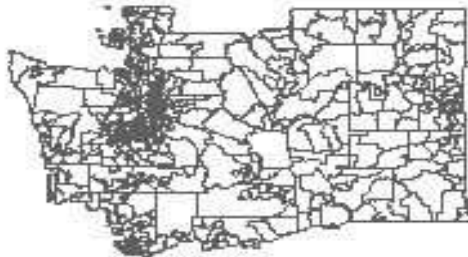
Spatial data types



cities • *Points*



railroads • *Lines*



zipcodes • *Polygons*

MINING SEQUENTIAL PATTERN IN TRANSACTIONAL DATABASES

SPATIAL DATA WAREHOUSING

SPATIAL DATA WAREHOUSE

- ▣ A spatial data warehouse is a specialized database that stores and manages spatial data. It's like a regular data warehouse, but specifically designed to handle geographic information.

SPATIAL DATACUBE

- ▣ It's a way to organize and analyze spatial data in a multidimensional space.
- ▣ it as a cube with multiple dimensions, where each dimension represents a different attribute of the data, such as time, location, and other variables.

DIMENSIONS AND MEASURES IN SPATIAL DATA WAREHOUSE

- ▣ **Dimensions** represent the different attributes or characteristics of the spatial data, such as time, location, or any other relevant variables. They provide the context or perspective for analyzing the data.
- ▣ On the other hand, **measures** are the quantitative values or metrics associated with the spatial data. These can be things like population density, temperature, pollution levels, or any other numerical values that you want to analyze.
- ▣ By combining dimensions and measures, you can perform various analytical operations, such as aggregating data based on different dimensions or comparing measures across different dimensions.
- ▣ This helps you gain insights and understand the relationships between different attributes of the spatial data.
- ▣ So, dimensions provide the context, while measures provide the quantitative values for analysis in a spatial data warehouse. It's like putting all the pieces together to understand the bigger picture.

STEPS INVOLVED IN SPATIAL MINING ANALYSIS

- ▣ spatial mining analysis typically involves the following steps:

1. Data Collection:

Gather spatial data from various sources, such as satellite imagery, GPS data, sensor networks, or public databases. Ensure the data is relevant to the analysis objectives.

2. Data Preprocessing:

Clean and preprocess the spatial data to address issues like missing values, outliers, and inconsistencies. This may involve data cleaning, transformation, normalization, and spatial referencing.

3. Spatial Data Integration:

If working with multiple datasets, integrate them into a unified spatial dataset. Ensure compatibility in terms of coordinate systems, resolution, and data structures.

4. Spatial Data Exploration:

Explore the spatial data to understand its characteristics, distributions, and relationships. Use visualizations, descriptive statistics, and exploratory data analysis techniques to gain insights.

5. Spatial Pattern Detection:

Apply spatial mining algorithms and techniques to identify patterns, trends, and relationships in the data. This may involve methods like clustering, classification, association rule mining, or spatial regression.

6. Pattern Evaluation:

Assess the discovered patterns for their significance, reliability, and usefulness. Evaluate the results against domain knowledge, statistical tests, or validation techniques.

7. Spatial Visualization:

Visualize the results using maps, charts, or interactive visualizations to communicate the spatial patterns effectively. This helps in understanding and interpreting the findings.

8. Interpretation and Decision-Making:

Analyze the patterns in the context of the problem domain and make informed decisions or recommendations based on the insights gained from the spatial mining analysis.

Spatial Data Mining Techniques:

- **Clustering:** Identifying groups of spatially close entities that share similar characteristics. This can be useful in identifying spatial patterns or groupings in a dataset.
- **Classification:** Assigning predefined categories or labels to spatial objects based on their attributes or spatial relationships. This is often used for land cover classification, for example.
- **Spatial Association Rule Mining:** Discovering relationships between spatially co-located features. For instance, identifying that certain types of businesses tend to be located near each other.
- **Spatial Outlier Detection:** Identifying spatial objects that deviate significantly from the expected patterns. This is useful in detecting anomalies or unusual occurrences in spatial data.
- **Spatial Regression:** Analyzing the relationships between spatial variables and predicting one variable based on others. It is an extension of traditional regression analysis to spatial data.

APPLICATIONS

1. Urban Planning:

Spatial mining can be used to analyze urban development patterns, identify areas of high population density, and determine optimal locations for infrastructure development such as schools, hospitals, or transportation networks.

2. Environmental Monitoring:

Spatial mining techniques can help analyze and predict patterns of environmental phenomena like air pollution, water quality, or deforestation. This information can aid in decision-making for environmental management and conservation efforts.

3. Public Health:

Spatial mining can assist in tracking the spread of diseases, identifying disease hotspots, and predicting potential outbreak areas. It can also help in optimizing the allocation of healthcare resources and planning vaccination campaigns.

4. Transportation and Logistics:

Spatial mining can optimize route planning, fleet management, and supply chain operations. It can help identify traffic patterns, congestion areas, and optimal locations for warehouses or distribution centers.

5. Retail and Marketing:

Spatial mining can analyze customer behavior, identify market segments, and optimize store locations. It can also assist in targeted advertising, personalized recommendations, and understanding consumer preferences based on their spatial patterns.

6. Natural Disaster Management:

Spatial mining can aid in predicting and mitigating the impact of natural disasters such as hurricanes, earthquakes, or floods. It can help identify vulnerable areas, plan evacuation routes, and allocate resources for emergency response.

CONCLUSION

- In conclusion, spatial data mining offers numerous advantages in analyzing spatial data, such as discovering hidden patterns, enhancing decision-making, optimizing resource allocation, and improving risk assessment.
- It provides a valuable tool for understanding spatial relationships, identifying trends, and making informed decisions in various domains.
- However, it is important to consider the potential challenges, including data complexity, data quality issues, privacy concerns, interpretation challenges, computational complexity, and selection bias.
- By addressing these challenges and leveraging the benefits of spatial data mining, we can unlock valuable insights and drive meaningful outcomes in spatial analysis.

THANK YOU!!

Sequence pattern mining

- ▣ A sequence database consists of sequences of ordered elements or events, recorded with or without a concrete notion of time. the sequence data such as customer shopping sequences, web clickstreams etc.
- ▣ Sequential pattern mining is the mining of frequently occurring ordered events or subsequences as patterns.
- ▣ For retail data, sequential patterns are useful for shelf placement and promotions. this industry, as well as telecommunications and other businesses, may also use sequential patterns for targeted marketing, customer retention, and many other tasks. other areas in which sequential patterns can be applied include web access pattern analysis, weather prediction, production processes, and network intrusion detection.
- ▣ Transactional databases are for critical business transactions where high level of data integrity is required ex: banking.

- Given a set of sequences, where each sequence consists of a list of events (or elements) and each event consists of a set of items, and given a user-specified minimum support threshold of min sup, sequential pattern mining finds all frequent subsequences, that is, the subsequences whose occurrence frequency in the set of sequences is no less than min sup.
- Instead of writing individual items and their transaction id s we group the items and generate sequences from them so that it will be easy to analyze .

□ For eg: TID items

1 a,b,c,d

2 a,f,d,c

3 d,e,f

4 a,d,e

→

sno

sequence

1

((abcd),(def))

2

((afdc),(ade))

- a sequence is formally defined as the ordered set of items $\{s_1, s_2, s_3, \dots, s_n\}$. as the name suggests, it is the sequence of items occurring together. it can be considered as a transaction or purchased items together in a basket.
- **subsequence:** the subset of the sequence is called a subsequence. suppose $\{a, b, g, q, y, e, c\}$ is a sequence. the subsequence of this can be $\{a, b, c\}$ or $\{y, e\}$. observe that the subsequence is not necessarily consecutive items of the sequence. from the sequences of databases, subsequences are found from which the generalized sequence patterns are found at the end.
- ▣ Our task is to mine such subsequences that are frequently occurring in the given database.

Scalable methods for mining

- ▣ Sequential pattern mining is computationally challenging because such mining may generate and/or test a combinatorially explosive number of intermediate subsequences.
- ▣ We can adopt two ways to achieve this: (1) efficient methods for mining the full set of sequential patterns, and (2) efficient methods for mining only the set of closed sequential patterns, because all of the subsequences of a frequent sequence are also frequent, mining the set of closed sequential patterns may avoid the generation of unnecessary subsequences and thus lead to more compact results as well as more efficient methods than mining the full set.
- ▣ There are three such approaches and all three approaches either directly or indirectly explore the apriori property.

Scalable methods for mining

- ▣ The following are the methods for mining:
 - 1) GSP (Generalize Sequential Patterns)
 - 2) SPADE
 - 3) Prefix Span: Prefix-Projected Sequential Pattern Growth

GSP (Generalize Sequential Patterns)

- GSP is a very important algorithm in data mining. It is used in sequence mining from large databases. This sequence mining algorithm is basically based on apriori algorithm.
- GSP uses a level-wise paradigm for finding all the sequence patterns in the data. It starts with finding the frequent items of size one and then passes that as input to the next iteration of the GSP algorithm. The database is passed multiple times to this algorithm. In each iteration, it removes all the non-frequent item sets.
- This is done based on a threshold frequency which is called support. Only those item sets are kept whose frequency is greater than the support count. After the first pass, GSP finds all the frequent sequences of length-1 which are called 1-sequences. This makes the input to the next pass, it is the candidate for 2-sequences. At the end of this pass, GSP generates all frequent 2-sequences, which makes the input for candidate 3-sequences. The algorithm is recursively called until no more frequent item sets are found.

EXAMPLE:

SID	seq
100.	((a,b),(c),(f,g),(g),(e))
200.	((a,d),(c),(b),(a,b,e,f))
300.	((a),(b),(f,g),(e))
400.	((b),(f,g))

- *< is a sequence whereas are the elements of the sequence. these elements are sometimes referred as transactions. an element may contain a set of items. items within an element are unordered and we list them alphabetically.*
- *for example, (cef) is the element and it consists of 3 items c, e and f.*
- *Let minimum support=2.*
- *The 1-length sequences possible are*

1-len	Support
(a)	3
(b)	4
(c)	2
(d)	1
(e)	3
(f)	4
(g)	3

- Join Phase: we generate the candidate key by joining L_{k-1} and L_{k-1} . while joining we need to add the item in s_2 both as part of an itemset and separate element, since both $((x)(y))$ and (xy) give the same sequence (y) upon deleting the first item.
- Prune Phase: we delete candidate sequences that have a contiguous $(k-1)$ sub sequences whose support count is less than minimum support.

2-len seq (a)	2-len seq (b)	2-len seq(c)	2-len seq(d)	2-len seq(e)	2-len seq(f)
(ab) 2	(ba) 1	(ca) 1	(ea) 0	(fa) 0	(ga) 0
(ac) 2	(bc) 1	(cb) 1	(eb) 0	(fb) 0	(gb) 0
(ae) 3	(be) 3	(ce) 2	(ec) 0	(fc) 0	(gc) 0
(af) 3	(bf) 4	(cf) 2	(ef) 0	(fg) 2	(ge) 2
(ag) 2	(bg) 3	(cg) 1	(eg) 0	(fe) 1	(gf) 0
(a,b) 2	(b,c) 0	(c,e) 0	(e,f) 1	(f,g) 3	
(a,c) 0	(b,e) 1	(c,f) 0	(e,g) 1		
(a,e) 1	(b,f) 1	(c,g) 0			
(a,f) 1	(b,g) 1				
(a,g) 0					

▣ To generate a three length sequence we need to check that in two 2-len sequences after removing the left most and right most item we get the same sequence. The generated 3-len sequence elements should all be there in 2-len.

3-len seq	
(abe)	((ab)f)
(abf)	((ab)g)
(abg)	(bfe)
(ace)	(b(f,g))
(acf)	(cfe)
(afe)	(c(f,g))
(a(f,g))	((f,g)e)

2-len	Seq
(ab)	2
(ac)	2
(ae)	3
(af)	3
(ag)	2
(a,b)	2
(be)	3
(bf)	4
(bg)	3
(ce)	2
(cf)	2
(fe)	2
(f,g)	3
(g,e)	2

- ▣ Based on the condition prev discussed we remove $(c(f,g))$,we get

3-len seq			
(abe)	2	((ab)e)	1
(abf)	2	((ab)f)	1
(abg)	1	((ab)g)	1
(ace)	2	(bfe)	2
(acf)	2	(b(f,g))	3
(afe)	2	(bge)	2
(a(f,g))	2	(cfe)	1
(age)	2	((f,g)e)	2

□ The candidate key generated is

□ In the same way we generate l4 i.e;

3-len seq	support
(abe)	2
(abf)	2
(ace)	2
(acf)	2
(afe)	2
(a(f,g))	2
(age)	2
(bfe)	2
(b(f,g))	3
(bge)	2
((f,g)e)	2

4-len seq	Support
(abfe)	2
(ab(f,g))	1
(a(f,g)e)	2
(b(f,g)e)	2

- the candidate key for l4 is

4-len	Support
(abfe)	2
(a(f,g)e)	2
(b(f,g)e)	2

- In this way we generate the sequences for a given database.

Applications

1. **fraud detection:** gsp mining can be used to identify patterns of behavior that are indicative of fraud, such as unusual patterns of transactions or access to sensitive data.
2. **website navigation:** gsp mining can be used to analyze website navigation patterns, such as the sequence of pages visited by users, and identify areas of the website that are frequently accessed or ignored.
3. **sensor data analysis:** gsp mining can be used to analyze sensor data, such as data from iot devices, and identify patterns in the data that are indicative of certain conditions or states.
4. **social media analysis:** gsp mining can be used to analyze social media data, such as posts and comments, and identify patterns in the data that indicate trends, sentiment, or other insights.
5. **medical data analysis:** gsp mining can be used to analyze medical data, such as patient records, and identify patterns in the data that are indicative of certain health conditions or trends.

TEXT MINING

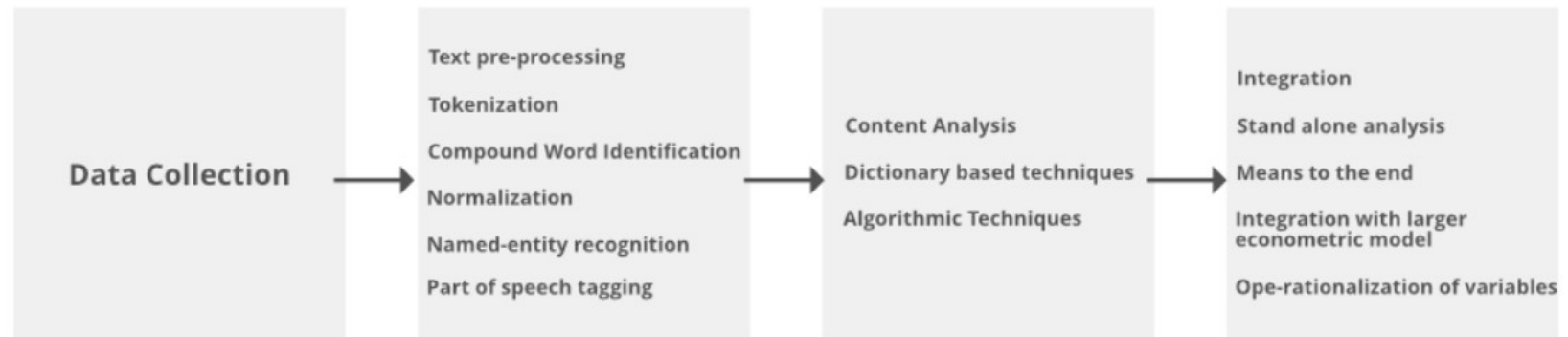
INTRODUCTION

Text mining is a component of data mining that deals specifically with unstructured text data. It involves the use of natural language processing (NLP) techniques to extract useful information and insights from large amounts of unstructured text data. Text mining can be used as a preprocessing step for data mining or as a standalone process for specific tasks.

By using text mining, the unstructured text data can be transformed into structured data that can be used for data mining tasks such as classification, clustering, and association rule mining. This allows organizations to gain insights from a wide range of data sources, such as customer feedback, social media posts, and news articles.

CONVENTIONAL PROCESS

- Gathering unstructured information from various sources accessible in various document organizations, for example, plain text, web pages, PDF records, etc.
- Pre-processing and data cleansing tasks are performed to distinguish and eliminate inconsistency in the data. The data cleansing process makes sure to capture the genuine text, and it is performed to eliminate stop words stemming (the process of identifying the root of a certain word and indexing the data).
- Processing and controlling tasks are applied to review and further clean the data set.
- Pattern analysis is implemented in Management Information System.
- Information processed in the above steps is utilized to extract important and applicable data for a powerful and convenient decision-making process and trend analysis.



TECHNIQUES

➤ **Information Retrieval** : In this process, we try to process the available documents and the text data into a structured form so, that we can apply different pattern recognition and analytical processes. It is a process of extracting relevant and associated patterns according to a given set of words or text documents. For this, we have processes like Tokenization of the document or the stemming process in which we try to extract the base word or the root word present there.

➤ **Information Extraction** : It is a process of extracting meaningful words from documents.

Feature Extraction - In this process, we try to develop some new features from existing ones. This objective can be achieved by parsing an existing feature or combining two or more features based on some mathematical operation.

Feature Selection - In this process, we try to reduce the dimensionality of the dataset which is generally a common issue while dealing with the text data by selecting a subset of features from the whole dataset.

Natural Language Processing :

Natural Language Processing includes tasks that are accomplished by using Machine Learning and Deep Learning methodologies. It concerns the automatic processing and analysis of unstructured text information.

Named Entity Recognition - Identifying and classifying named entities such as people, organizations, and locations in text data.

Sentiment Analysis - Identifying and extracting the sentiment (e.g. positive, negative, neutral) of text data.

Text Summarization - Creating a condensed version of a text document that captures the main points.

APPLICATIONS

- **Digital library** : A useful place to find digital text data is a library. It provides a new method of obtaining meaningful data in a way that allows millions of records to be accessed online. A robust approach for extracting reports that handle several forms, such as Microsoft Word, PDF, HTML, scripting languages.
- **Academic and Research Field** : Various text-mining techniques and tools are applied in the field of education to investigate the instructional patterns within a certain area of study. The primary goal of using text mining in research is to assist in finding and organizing relevant data and research papers from multiple domains on a single platform. To do this, we employ k-Means clustering, and other techniques.

Contd..

- **Life science:** Life science and healthcare industries are producing an enormous volume of textual and mathematical data regarding patient records, sicknesses, medicines, symptoms, and treatments of diseases, etc. It is a major issue to filter data and relevant text to make decisions from a biological data repository.
- **Social-Media:** Text mining can be used to break down and examine web-based media applications, such as emails, blogs, web journals, and internet news, in order to track and look into online information. Text mining tools facilitate the identification and analysis of the quantity of posts, likes, and followers on a social media platform.
- **Resume Filtering:** Each day, large companies and job agencies receive hundreds of thousands of applications from job seekers. It is difficult to extract data from resumes with good recall and precision. The first step for filtering resumes might be automated information extraction. Consequently, it is crucial to automate the resume screening process.

ADVANTAGES

- ❑ **Variety of Applications:** Text mining has a wide range of applications, including sentiment analysis, named entity recognition, and topic modeling. This makes it a versatile tool for organizations to gain insights from unstructured text data.
- ❑ **Large Amounts of Data:** Text mining allows organizations to extract insights from large amounts of unstructured text data. This can include customer feedback, social media posts, and news articles.
- ❑ **Improved Decision Making:** Text mining can be used to extract insights from unstructured text data, which can be used to make data-driven decisions.
- ❑ **Cost-effective:** Text mining can be a cost-effective way to extract insights from unstructured text data, as it eliminates the need for manual data entry.
- ❑ **Broader benefits:** Cost reductions, productivity increases, the creation of novel new services, and new business models.

DISADVANTAGES

- ❑ **Complexity:** Text mining can be a complex process that requires advanced skills in natural language processing and machine learning.
- ❑ **Quality of Data:** The quality of text data can vary, which can affect the accuracy of the insights extracted from text mining.
- ❑ **High Computational Cost:** Text mining requires high computational resources, and it may be difficult for smaller organizations to afford the technology.
- ❑ **Limited to Text Data:** Text mining is limited to extracting insights from unstructured text data and cannot be used with other data types.
- ❑ **Noise in text mining results:** Text mining of documents may result in mistakes. It's possible to find false links or to miss others. In most situations, if the noise (error rate) is sufficiently low, the benefits of automation exceed the chance of a larger mistake than that produced by a human reader.

CONCLUSION

Text mining is an effective tool for identifying trends and insights in text data and has many applications. It can be enhanced by combining it with other techniques, such as natural language processing and machine learning. Text mining identifies relevant information within a text and therefore, provides qualitative results.

Overall, it is an important tool for extracting insights from text data that can be used to inform decision-making and improve business outcomes.

THANK YOU

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MULTIMEDIA DATA MINING

INTRODUCTION

Multimedia data mining is like searching for hidden treasures in a mix of different media like pictures, videos, sounds, and text. It's about using computer techniques to find patterns, interesting information, or connections in this diverse pile of data. This helps us make sense of large amounts of information and discover useful things, like trends or insights, in the world of multimedia.

CLASSIFICATION OF DATA MINING:

- 1.Image Mining:** Analyzing patterns and extracting information from images.
- 2.Video Mining:** Exploring patterns and knowledge within video data, often involving scene analysis and motion detection.
- 3.Audio Mining:** Extracting meaningful patterns or information from audio sources, such as speech recognition or music analysis.
- 4.Text and Image Integration:** Combining textual and visual information for comprehensive analysis, commonly used in applications like image captioning.

5.Cross-Media Mining: Extracting knowledge from multiple types of media, exploring relationships and patterns across different modalities like images, text, and audio.

6.Content-Based Retrieval: Finding multimedia content based on its features or characteristics, such as searching for images similar to a given one.

7.Multimedia Data Clustering: Grouping similar multimedia data together based on certain characteristics or features.

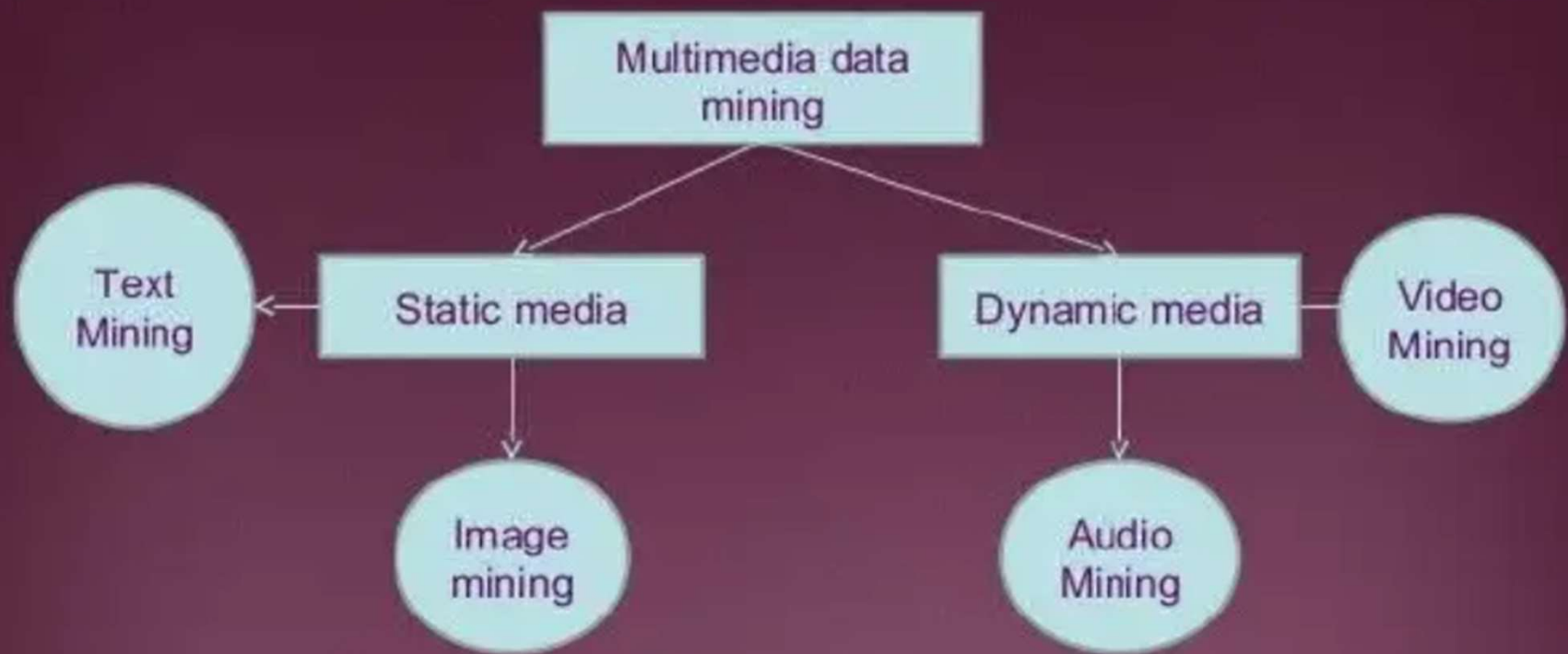


Fig. Categories of Multimedia Data Mining

MULTIMEDIA DATA MINING PROCESS:

Multimedia data mining involves the extraction of useful information, patterns, and knowledge from multimedia data, which can include text, images, audio, video, and other forms of media. The process of multimedia data mining typically involves several key steps:

1.Data Collection:

1. Gather multimedia data from various sources, such as databases, the internet, sensor networks, or other repositories.

2.Preprocessing:

1. Clean and preprocess the multimedia data to eliminate noise and irrelevant information.
2. For text data, this may involve tasks like stemming, stop-word removal, and normalization.

3.Feature Extraction:

1. Identify and extract relevant features from the multimedia data.
2. Features are characteristics or properties that represent the data in a more manageable and meaningful way.
3. In image data, features could be color histograms, texture patterns, or shape descriptors.

4.Data Representation:

1. Represent the multimedia data using appropriate data structures and formats for analysis.
2. This step involves transforming the extracted features into a format suitable for mining algorithms.

5.Data Mining Algorithms:

1. Apply data mining techniques to discover patterns, trends, associations, or clusters in the multimedia data.
2. Common data mining techniques include clustering, classification, association rule mining, and anomaly detection.

6. Pattern Evaluation:

1. Evaluate and interpret the discovered patterns for their significance and usefulness.
2. This step involves assessing the quality and relevance of the mined patterns.

7. Knowledge Discovery:

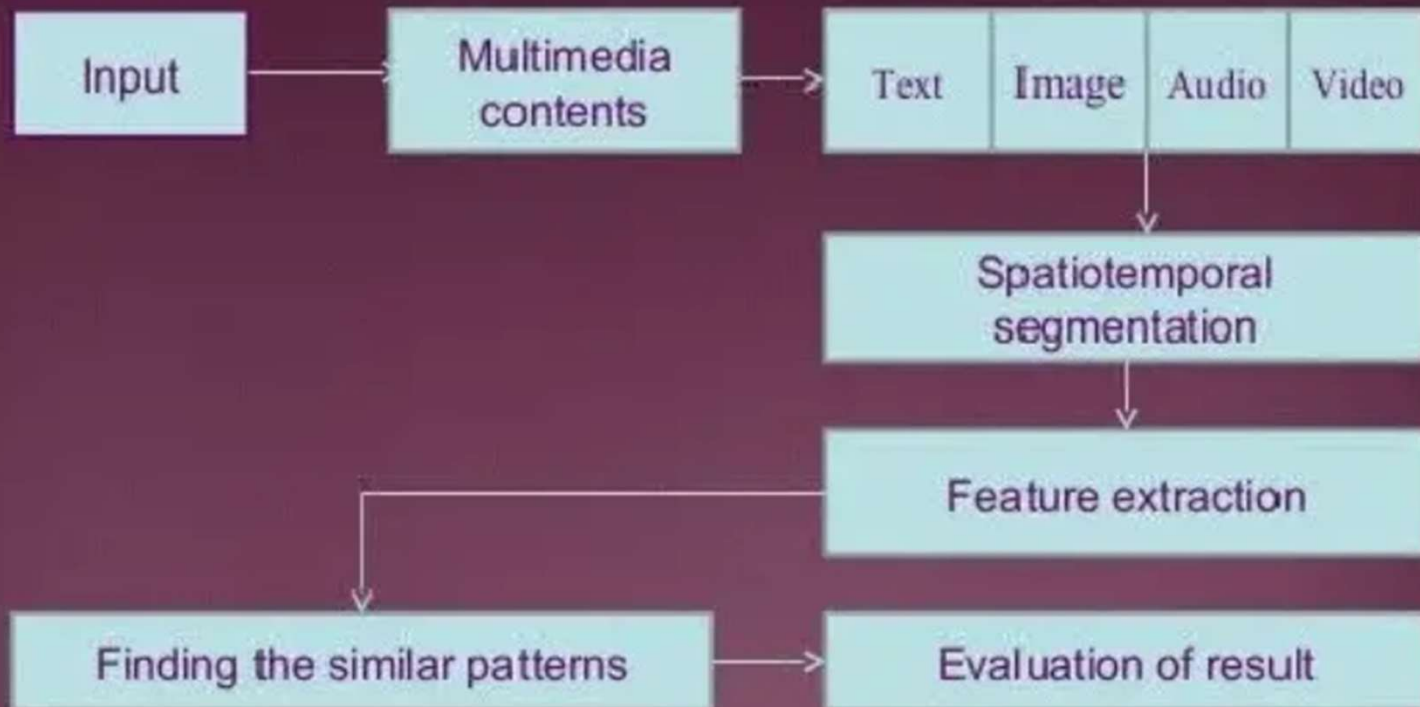
1. Derive meaningful knowledge from the evaluated patterns.
2. This knowledge can be used for decision-making, prediction, and other applications.

8. Visualization:

1. Present the mined knowledge in a visually understandable format.
2. Visualization aids in the interpretation and communication of the discovered patterns.

9. Post-Processing:

1. Refine and optimize the discovered knowledge if needed.
2. This step may involve further analysis or feedback loops to improve the mining results.



ALGORITHMS USED FOR MULTIMEDIA DATA MINING:

Various algorithms are employed for multimedia data mining, depending on the type of data and the specific goals of the analysis. Here are some commonly used algorithms across different multimedia domains:

1. Image and Video Analysis:

- SIFT (Scale-Invariant Feature Transform)
- SURF (Speeded-Up Robust Features)
- CNNs (Convolutional Neural Networks)

2. Audio Analysis:

- MFCC (Mel-Frequency Cepstral Coefficients)
- Hidden Markov Models (HMMs)

3.Text Analysis:

- TF-IDF (Term Frequency-Inverse Document Frequency)
- NLP Techniques (Natural Language Processing)

4.Cross-modal Analysis:

- Multimodal Deep Learning Models
- Canonical Correlation Analysis (CCA)

5.Clustering and Classification:

- K-Means Clustering:
- Decision Trees and Random Forests

6Association Rule Mining:

- Apriori Algorithm

ADVANTAGES OF MULTIMEDIA DATAMINING:

1. Rich Information Source
2. Improved Decision Making
3. Enhanced User Experience
4. Discovery of Complex Patterns
5. Cross Model Analysis
6. Applications in Various Fields

DISADVANTAGES OF MULTIMEDIA DATAMINING:

- 1.Data Complexity
- 2.Computational Intensity
- 3.Subjectivity in Interpretation
- 4.Privacy Concerns
- 5.Data Annotation Challenges
- 6.Limited Explainability
- 7.Semantic Gap

CONCLUSION:

In conclusion, multimedia data mining is a powerful and dynamic field with both notable advantages and challenges. The ability to extract insights from diverse data types, including text, images, audio, and video, provides a rich source of information for decision-making across various domains. The discovery of complex patterns and the potential for cross-modal analysis contribute to a more holistic understanding of multimedia content.

THANKYOU

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SOCIAL NETWORK ANALYSIS

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III CSE E

Topics covered :

- ▣ Introduction to Social Network Analysis
- ▣ Key components of Social Network Analysis
- ▣ Significance in Data warehousing and Datamining
- ▣ Integrating SNA into Data warehousing
- ▣ Leveraging SNA for Data Mining
- ▣ Applications
- ▣ Challenges and considerations
- ▣ Tools and Technologies

Social Network Analysis

- ▣ SNA is a qualitative and quantitative research method that studies relationships between individuals or entities within a network.
- ▣ Social Network Analysis (SNA) is a field of study that examines the relationships and interactions between individuals, groups, or organizations. It provides a framework for analyzing social structures, patterns, and dynamics by representing them as networks. These networks consist of nodes (representing entities) and edges (representing relationships or interactions between entities). SNA can be applied to various social structures, including friendships, collaborations, communication patterns, and more.
- ▣ The examples of Social Networks are Facebook, Twitter, Google+, or any other websites.



**SOCIAL
NETWORK**
.....

Key components of Social Network Analysis

- ▣ Nodes
- ▣ Edges
- ▣ Types of Networks
- ▣ Centrality Measures
- ▣ Cliques and Subgroups
- ▣ Network Density
- ▣ Path length
- ▣ Network Hierarchy
- ▣ Network Motifs

1)Nodes:

- Nodes represent individual entities in the network, such as people, organizations, or even concepts.
- In a social context, nodes could be individuals, and in a business context, nodes could be employees.

2)Edges:

- Edges, also known as ties or links, represent relationships or interactions between nodes.
- For example, in a social network, an edge could represent a friendship or a professional collaboration.
- The edges may be directed, undirected, weighted.

3) Types of Networks

- A network is the entire structure formed by nodes and edges.
- Social networks can be visualized and analyzed to understand patterns and behaviors.
- The networks may be directed, undirected, weighted.

4) Centrality measures

- Degree Centrality:** Focuses on the number of direct connections a node has.
- Betweenness Centrality:** Emphasizes nodes that act as bridges or intermediaries in the network.
- Eigenvector Centrality:** Considers both the quantity and quality of a node's connections, giving more weight to connections with other central nodes.

5)Cliques and Subgroups

- >A clique is a subset of nodes in a network where every node is directly connected to every other node within the subset. In simpler terms, it is a group of individuals who are all directly connected to each other.
- >Subgroups, or sub-communities, refer to smaller groups of nodes within a larger network that share a higher density of connections among themselves than with nodes outside the subgroup.

6) Network Density

Density is a measure of how closely knit a network is, reflecting the proportion of actual connections to possible connections.

- high density
- low density

7) Network Hierarchy

- reflects the arrangement of nodes in terms of power, influence or importance

8) Network Motifs

- Recurring and statistically significant subgraph patterns

Significance in Data warehousing and Data mining :

- ▣ Social Network Analysis (SNA) is highly significant in data warehousing and data mining as it enriches traditional datasets with social network information, providing a comprehensive view of relationships and communication dynamics. In data mining, SNA is a powerful tool for uncovering hidden patterns and identifying influential nodes within networks, contributing to more informed decision-making. This integration enhances the depth of insights derived from structured data, enabling businesses to optimize strategies and predict trends effectively.
- ▣ Application of advanced analytics to uncover valuable patterns, anomalies and trends.

Integrating SNA into Data warehousing

- ▣ Data Collection
- ▣ Schema Design
- ▣ Data Storage
- ▣ Data Integration
- ▣ Query and Analysis
- ▣ Security and privacy

1. Data Collection:

Data collection is the process of gathering information from various sources to be used for analysis, reporting, or decision-making.

In the context of SNA, data collection may extend to include social network data, such as information about relationships, interactions, and network structure. This could involve collecting data from social media platforms, communication channels, or collaboration tools.

2. Schema Design:

Schema design involves creating the structure for organizing and representing data in a database. It includes defining tables, relationships, and constraints.

In the integration of SNA, the schema design may include additional tables or structures to represent nodes, edges, and other attributes specific to social network data. This extended schema accommodates the unique characteristics of social relationships and interactions.

3. Data Storage:

Data storage refers to the physical or logical storage of collected and structured data in a database or other storage systems.

SNA data may be stored in dedicated tables or datasets designed to capture social network information. This could involve specialized storage mechanisms that optimize for graph-based data structures.

4. Data Integration:

Data integration involves combining data from different sources to provide a unified view, often through ETL (Extract, Transform, Load) processes.

Integrating SNA involves merging traditional data with social network data. ETL processes may include transformations specific to handling relationships, identifying influencers, and creating a cohesive network structure within the data warehouse.

5. Query and Analysis:

Query and analysis involve extracting insights and valuable information from the stored data using queries, reporting tools, and analytical techniques.

SNA extends query and analysis capabilities to include network analysis metrics, identifying clusters, central nodes, and patterns within the social network. Graph databases or specialized SNA algorithms may be employed for these analyses.

6. Security and Privacy:

Security and privacy involve protecting data from unauthorized access, ensuring data integrity, and safeguarding sensitive information.

Integrating SNA into data warehousing requires additional considerations for securing social network data. This may involve anonymization techniques for privacy, access controls specific to social network elements, and compliance with regulations related to social data.

Leveraging SNA for Data Mining

- ▣ Data collection and preprocessing
- ▣ Graph representation
- ▣ Centrality Analysis
- ▣ Community Detection
- ▣ Pattern Recognition
- ▣ Predictive Analysis
- ▣ Visualization
- ▣ Classification and Clustering
- ▣ Evaluation and validation
- ▣ Integration with Decision making

1) Data collection and preprocessing

-It ensures that the data is accurate, consistent, and ready for analysis.

2) Graph representation

- Graph representation is a way of expressing and visualizing relationships and connections between entities using a mathematical structure called a graph. In the context of SNA or data analysis in general, a graph represents a set of nodes (points) and edges (connections) between those nodes.

3) Centrality Analysis

-Centrality analysis is a key concept in SNA that helps identify the most important or central nodes within a network. It measures the relative importance of nodes based on their position and influence in the network structure.

4) Community Detection

- It involves identifying groups of nodes in a network that are more densely connected to each other than to nodes outside the group. These groups are often referred to as "communities" or "clusters."

The goal is to uncover the underlying structure and organization within a network.

5) Pattern Recognition

-Pattern recognition is the process of identifying regularities or patterns within data.

It involves the development of algorithms and models that can learn from data, recognize patterns, and make predictions or decisions based on those patterns.

6) Predictive Analysis

-Predictive analysis involves using data, statistical algorithms, and machine learning to identify the likelihood of future outcomes based on historical data. The goal is to make predictions about future events or trends.

7) Visualization

-Visualization in Social Network Analysis (SNA) involves graphically representing nodes and edges to reveal patterns, relationships, and structures within a network.

8) Classification and Clustering

-Classification assigns predefined labels to data based on patterns, while clustering groups similar data points without predefined labels.

9) Evaluation and validation

Applications:

- ▣ **Community Detection:**

Identifying clusters or communities of tightly connected nodes within a network.

- ▣ **Influence and Opinion Spread:**

Analyzing how information or influence spreads through a network.

- ▣ **Identifying Key Players:**

Determining influential nodes or individuals who play crucial roles in a network.

- ▣ **Collaboration and Communication:**

Understanding patterns of collaboration and communication within organizations.

- ▣ **Fraud Detection:**

Detecting anomalous patterns in networks that may indicate fraudulent activities.

Challenges and considerations

- ▣ **Data Quality and Reliability:**

Challenge: Incomplete, inaccurate, or biased data can lead to unreliable network representations.

enhance Consideration: Rigorous data validation processes and the use of multiple data sources to data quality.

- ▣ **Privacy Concerns:**

Challenge: Balancing the need for analysis with individuals' privacy rights.

obtaining Consideration: Implementing robust privacy policies, anonymization techniques, and informed consent.

- ▣ **Bias in Network Data:**

Challenge: Networks may not fully represent the complexities of real-world relationships.

interpretation of Consideration: Acknowledging and addressing biases, and considering context in the results.

Tools and Technologies

- ▣ Visualisation tools
 - Gephi, Cytoscape represent network structures Graphically
- ▣ Integration with Datamining tools
 - Datamining Tools, Weka, Rapidmines for comprehensive analysis
- ▣ Bigdata Technologies
 - Hadoop, Spark – Scalable SNA

CONCLUSION:

- ▣ Social Network Analysis is a powerful tool for understanding the structure and dynamics of social relationships.
- ▣ Integration with Data Warehouse and Data Mining enhances the depth and accuracy of analysis.

THANK YOU

MINING TIME-SERIES DATA

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introduction

- ▣ Mining time series data is the process of analyzing and extracting useful information from data that is collected over time or at regular intervals. Time series data can be found in various domains, such as finance, economics, medicine, engineering, and science.
- ▣ Some of the common tasks and applications of mining time series data are

Time series forecasting

- ▣ Time series forecasting is the process of analyzing time series data using statistics and modeling to make predictions and inform strategic decision-making. Time series data is data that is collected over time or at regular intervals, such as stock prices, weather, sales, etc. Time series forecasting involves finding patterns and trends in the historical data and extrapolating them into the future.
- ▣ Some of the common methods and techniques for time series forecasting are:
 - Exponential smoothing: This is a family of methods that use weighted averages of past observations to smooth out the fluctuations and noise in the data. The weights decay exponentially as the observations get older, giving more importance to recent data. Some examples of exponential smoothing methods are simple exponential smoothing, Holt's linear trend method, and Holt-Winters seasonal method.

Time series forecasting

- Autoregressive integrated moving average (ARIMA): This is a class of models that use past values and errors of the time series to predict future values. ARIMA models can capture various characteristics of the data, such as stationarity, seasonality, trend, and autocorrelation. ARIMA models are specified by three parameters: p , d , and q , which represent the order of the autoregressive, integrated, and moving average components, respectively.
- Neural networks: These are computational models that mimic the structure and function of biological neurons. Neural networks can learn complex nonlinear relationships from the data and generalize to unseen data. Neural networks can be used for time series forecasting by feeding the historical data as inputs and training the network to output the future values. Some examples of neural network architectures for time series forecasting are feedforward neural networks, recurrent neural networks, and long short-term memory networks.

Time series classification

- ▣ Time series classification is the task of assigning a label or category to a time series based on its features or patterns. Time series are sequences of data points that are ordered in time or at regular intervals, such as sensor readings, stock prices, or speech signals. Time series classification can be useful for various applications, such as anomaly detection, activity recognition, or medical diagnosis.
- ▣ There are different methods and techniques for time series classification, such as distance-based, feature-based, model-based, or deep learning approaches. Some of the common challenges and issues in time series classification are:
 - The high dimensionality and variability of time series data
 - The choice of an appropriate similarity or distance measure for comparing time series
 - The extraction or selection of relevant and discriminative features from time series
 - The handling of noise, outliers, missing values, or irregular sampling rates in time series
 - The scalability and efficiency of the algorithms for large and complex time series data sets

Time series clustering

- ▣ Time series clustering is the process of grouping time series data based on their similarity or dissimilarity. Time series data are sequences of data points that are ordered in time or at regular intervals, such as sensor readings, stock prices, or speech signals. Time series clustering can be useful for various applications, such as anomaly detection, activity recognition, or pattern discovery.
- ▣ There are different methods and techniques for time series clustering, such as distance-based, feature-based, model-based, or deep learning approaches. Some of the common challenges and issues in time series clustering are:
 - The high dimensionality and variability of time series data
 - The choice of an appropriate similarity or distance measure for comparing time series
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 - The scalability and efficiency of the algorithms for large and complex time series data sets

Time series segmentation

- ▣ Time series segmentation is the process of dividing a time series into smaller segments or subseries based on some criteria or change points. Time series segmentation can be useful for various applications, such as anomaly detection, pattern discovery, feature extraction, or compression. Time series segmentation can be done using different methods and techniques, such as distance-based, model-based, or deep learning approaches. Some of the common challenges and issues in time series segmentation are:
 - The choice of an appropriate similarity or distance measure for comparing time series segments
 - The determination of the optimal number and location of the segment boundaries
 - The handling of noise, outliers, missing values, or irregular sampling rates in time series
 - The scalability and efficiency of the algorithms for large and complex time series data sets

Time series indexing

- ▣ Time series indexing is the process of creating a data structure or a representation that allows fast and efficient retrieval or comparison of time series data. Time series data are sequences of data points that are ordered in time or at regular intervals, such as sensor readings, stock prices, or speech signals. Time series indexing can be useful for various applications, such as searching, clustering, classification, or anomaly detection.
- ▣ There are different methods and techniques for time series indexing, such as distance-based, feature-based, model-based, or symbolic approaches. Some of the common challenges and issues in time series indexing are:
 - The high dimensionality and variability of time series data
 - The choice of an appropriate similarity or distance measure for comparing time series
 - The extraction or selection of relevant and discriminative features from time series
 - The handling of noise, outliers, missing values, or irregular sampling rates in time series
 - The scalability and efficiency of the algorithms for large and complex time series data sets

Time series rule discovery

- ▣ Time series rule discovery is the process of finding or generating rules or patterns that describe the behavior or relationship of time series data. Time series data are sequences of data points that are ordered in time or at regular intervals, such as sensor readings, stock prices, or speech signals. Time series rule discovery can be useful for various applications, such as prediction, anomaly detection, or pattern discovery.
- ▣ There are different methods and techniques for time series rule discovery, such as statistical, machine learning, deep learning, or symbolic approaches. Some of the common challenges and issues in time series rule discovery are:
 - The high dimensionality and variability of time series data
 - The choice of an appropriate similarity or distance measure for comparing time series
 - The extraction or selection of relevant and discriminative features from time series
 - The handling of noise, outliers, missing values, or irregular sampling rates in time series
 - The scalability and efficiency of the algorithms for large and complex time series data sets

Time series anomaly detection

- ▣ Time series anomaly detection is the process of identifying data points or patterns in a time series that deviate significantly from the norm. A time series is a collection of data points that are ordered in time or at regular intervals, such as stock prices, weather, sales, etc. Time series anomaly detection can be useful for various applications, such as fraud detection, intrusion detection, failure detection, or event detection.
- ▣ There are different methods and techniques for time series anomaly detection, such as statistical, machine learning, deep learning, or symbolic approaches. Some of the common challenges and issues in time series anomaly detection are:
 - The high dimensionality and variability of time series data
 - The choice of an appropriate similarity or distance measure for comparing time series
 - The extraction or selection of relevant and discriminative features from time series
 - The handling of noise, outliers, missing values, or irregular sampling rates in time series
 - The scalability and efficiency of the algorithms for large and complex time series data sets

Thank
you

MINING WORLD WIDE WEB

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DATA MINING- WORLD WIDE WEB

- Over the last few years, the **World Wide Web** has become a significant source of information and simultaneously a popular platform for business. Web mining can define as the method of utilizing data mining techniques and algorithms to extract useful information directly from the web, such as Web documents and services, hyperlinks, Web content, and server logs.



WHAT IS WEB MINING?

Web mining can widely be seen as the application of adapted data mining techniques to the web, whereas data mining is defined as the application of the algorithm to discover patterns on mostly structured data embedded into a **knowledge discovery process**.



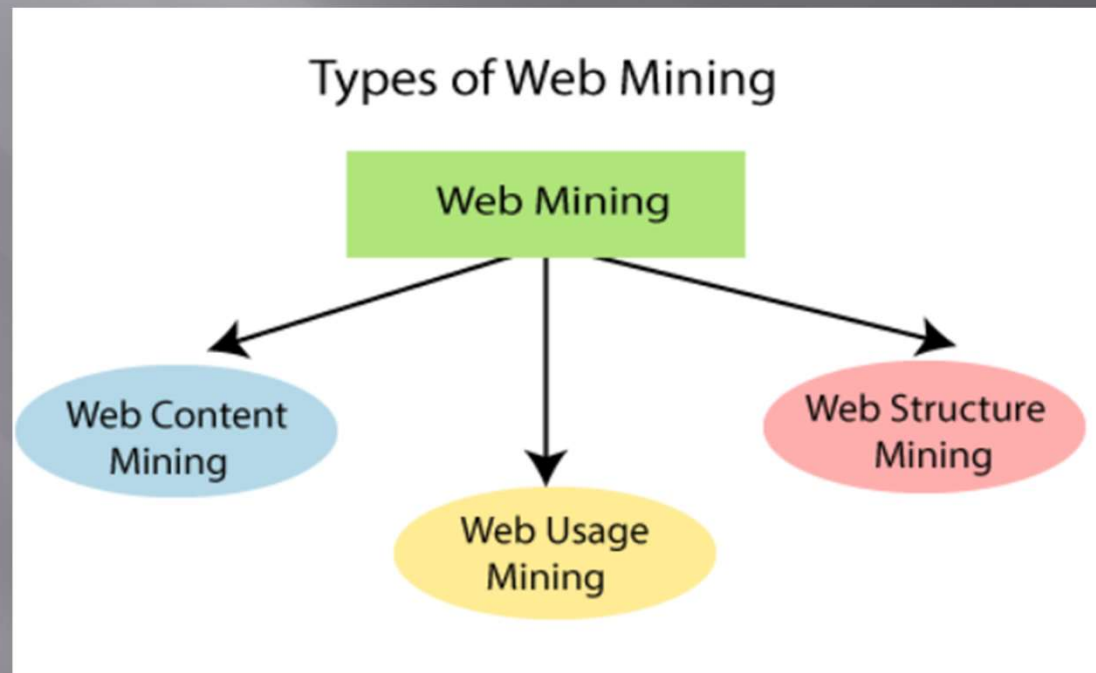
WHY MINE THE WEB?

- ▣ Enormous wealth of information on web
- ▣ Lots of data on user access patterns
- ▣ Possible to mine interesting nuggets of information

GOALS OF WEB MINING

- **Pattern Discovery:** Identifying interesting patterns, trends, and associations within web data.
- **Personalization:** Customizing content and services based on user preferences and behavior.
- **Recommendation Systems:** Recommending products, services, or content based on user profiles and behavior.
- **Search Engine Optimization (SEO):** Improving the visibility and ranking of websites in search engine results.
- **Fraud Detection:** Identifying fraudulent activities, scams, or malicious behavior on the web.

THERE ARE THREE TYPES OF WEB MINING



WEB CONTENT MINING

- ▣ Web content mining is a branch of web mining that focuses on extracting valuable information, patterns, and knowledge from the textual, visual, and multimedia content present on web pages.
- ▣ The primary objective is to analyze and understand the content of web pages, uncover hidden patterns, and derive meaningful insights.

OBJECTIVES OF WEB CONTENT MINING:

Information Extraction:

Identify and extract specific information from web pages. This may include names, entities, events, and other structured data.

Text Mining:

Analyze and process the textual content of web pages. This involves techniques such as natural language processing (NLP) and text analytics.

Sentiment Analysis:

Determine the sentiment or opinion expressed in the text, such as positive, negative, or neutral. This is often used in social media and customer reviews.

Content Categorization:

Categorize web content into predefined categories or topics. This helps in organizing and structuring information for easier retrieval.

Information Retrieval:

Enhance the efficiency of information retrieval by indexing and searching relevant content based on user queries.

WEB STRUCTURED MINING

- ▣ The web structure mining can be used to find the link structure of hyperlink. It is used to identify that data either link the web pages or direct link network.
- ▣ In Web Structure Mining, an individual considers the web as a directed graph, with the web pages being the vertices that are associated with hyperlinks.
- ▣ The most important application in this regard is the Google search engine, which estimates the ranking of its outcomes primarily with the PageRank algorithm

OBJECTIVES OF WEB STRUCTURE MINING

Topology Analysis:

Study the overall structure and topology of websites to identify key pages, hubs, and authorities. This helps in understanding the organization of information on the web.

Link Analysis:

Analyze the links between web pages to uncover patterns of hyperlinking behavior. This includes inbound links, outbound links, and their significance.

Page Ranking:

Assign importance scores or ranks to web pages based on their link structures. Algorithms like PageRank are commonly used to determine the importance of pages within a website.

Community Detection:

Identify communities or clusters of related pages within the web. Pages within a community are more strongly connected to each other than to pages outside the community.

WEB USAGE MINING

- ▣ Web usage mining is used to extract useful data, information, knowledge from the weblog records, and assists in recognizing the user access patterns for web pages.
- ▣ In Mining, the usage of web resources, the individual is thinking about records of requests of visitors of a website, that are often collected as web server logs.

OBJECTIVES OF WEB USAGE MINING

User Behavior Analysis:

Understand how users interact with websites, including page views, clicks, and navigation paths.

Pattern Discovery:

Identify recurring patterns in user behavior to gain insights into common navigation sequences and preferences.

Personalization:

Customize content and services based on individual user preferences and historical interactions.

Recommendation Systems:

Develop systems that recommend products, services, or content to users based on their browsing history and preferences.

User Profiling:

Build profiles of users based on their behavior, preferences, and demographics

CHALLENGES IN WEB MINING

- **The complexity of web pages:**

The site pages don't have a unifying structure. They are extremely complicated as compared to traditional text documents. There are enormous amounts of documents in the digital library of the web. These libraries are not organized according to a specific order.

- **The web is a dynamic data source:**

The data on the internet is quickly updated. For example, news, climate, shopping, financial news, sports, and so on.

- **Diversity of client networks:**

The client network on the web is quickly expanding. These clients have different interests, backgrounds, and usage purposes. There are over a hundred million workstations that are associated with the internet and still increasing tremendously.

- **Relevancy of data:**

It is considered that a specific person is generally concerned about a small portion of the web, while the rest of the segment of the web contains the data that is not familiar to the user and may lead to unwanted results.

- **The web is too broad:**

The size of the web is tremendous and rapidly increasing. It appears that the web is too huge for data warehousing and data mining.

IS WEB MINING ETHICAL?

- ▣ User privacy
- ▣ Transparency
- ▣ Data security
- ▣ Purpose and intention
- ▣ Avoiding harm
- ▣ Data ownership

Application of Web Mining

Web mining has an extensive application because of various uses of the web. The list of some applications of web mining is given below.

- Marketing and conversion tool
- Data analysis on website and application accomplishment.
- Audience behavior analysis
- Advertising and campaign accomplishment analysis.
- Testing and analysis of a site.

Thank you